Small Lyapunov ratios and the structure of Berry–Esseen maximizers

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6 October 2026

Abstract

For changing finite independent centered arrays whose Lyapunov ratio tends to zero and whose Berry–Esseen ratio tends to Esseen's constant, we prove approximation by coordinatewise laws on one common lattice. The sum of their squared and cubed Wasserstein errors is smaller than the corresponding lattice scale, and the original additive zero-bias transport bound is asymptotically an equality. If each row is an actual global maximizer at its own finite horizon, then every original representation is eventually exactly an IID two-point array padded by zeros. The resulting Lyapunov threshold is qualitative. The proof retains the absolute third moments of the original coordinates and uses a relative conditional first-moment estimate to exclude phase noise of arbitrarily small size.

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1 The results

Let $X_1, \dots, X_N$ be independent centered real variables with finite third absolute moments. Write

$$S = \sum_i X_i, \quad B^2 = \sum_i \mathbb{E}X_i^2 > 0, \quad R = \sum_i \mathbb{E}|X_i|^3, \quad L = R/B^3, \quad C = L^{-1} d_K(S/B, Z), \tag{1.1}$$

where Z is standard normal. All sums of third moments below refer to these original coordinates. Centered zero coordinates are allowed. Set

$$C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}}, \quad p_* = \frac{4-\sqrt{10}}{2}, \quad q_* = 1-p_*, \quad t_* = q_* - p_* = \sqrt{10} - 3,$$

$$r_* = p_*^2 + q_*^2, \quad \beta_* = \frac{r_*}{\sqrt{p_* q_*}}, \quad \nu_* = r_*^{-1}.$$

The variance-one Esseen law W is specified by $\mathbb{P}(W = \sqrt{q_*/p_*}) = p_*$ and $\mathbb{P}(W = -\sqrt{p_*/q_*}) = q_*$. For laws with finite jth moments, W_j denotes Wasserstein distance, with $W_j^j(X,Y) = \inf \mathbb{E}|\widetilde{X} - \widetilde{Y}|^j$, where the infimum is over couplings with the stated marginal laws. The zero-biased law S^* is defined by $\mathbb{E}[Sf(S)] = B^2 \mathbb{E}f'(S^*)$. The independent-sum zero-bias construction and the sharp single-law bound give

$$\mathcal{B}(X) := \frac{2B^2 W_1(S, S^*)}{R} \le 1; \tag{1.2}$$

see Goldstein [7], Goldstein and Reinert [8], Tyurin [11] and the equality classification in Bell [4, Theorem 1.1].

Theorem 1.1 (Changing arrays near Esseen's constant)**.** *In each row n, take a finite array satisfying (1.1). Its length may change. Suppose*

$$L_n \longrightarrow 0, \qquad C_n \longrightarrow C_{\mathrm{E}}. \tag{1.3}$$

There are positive gaps $h_n \sim R_n/(r_* B_n^2)$ and centered laws X_{ni}^{lat} on individual cosets of $h_n\mathbb{Z}$, coupled independently coordinate by coordinate to the original laws, such that

$$\sum_i W_2^2(X_{ni}, X_{ni}^{\mathrm{lat}}) = o(h_n^2), \qquad \sum_i W_3^3(X_{ni}, X_{ni}^{\mathrm{lat}}) = o(h_n^3). \tag{1.4}$$

Moreover $\mathcal{B}(X_n) \to 1$.

There also exist subsets I_n of active original coordinates, $m_n = |I_n|$, and a common sign within each row, such that I_n retains $1 - o(1)$ of both total variance and original price, $m_n L_n^2/\beta_*^2 \to 1$, and the padded IID laws

$$Q_{ni} = \mathrm{Law}(B_n W/\sqrt{m_n}) \quad (i \in I_n), \qquad Q_{ni} = \delta_0 \quad (i \notin I_n)$$

obey

$$\sum_i W_2^2(X_{ni}, Q_{ni}) = o(B_n^2), \qquad \sum_i W_3^3(X_{ni}, Q_{ni}) = o(R_n).$$

Here W is the variance-one Esseen two-point law or its reflection. The total variance of the target is exactly B_n^2 and its original price is asymptotic to R_n.

The lattice comparators in (1.4) may have more than two atoms. Their exact coordinate variances and cubic prices can differ from those of the originals. The IID approximation has a weaker error scale. A rare integer excursion can have a negligible contribution relative to R_n while preventing a binary approximation at scale h_n.

Let C_N be the supremum of (1.1) over all arrays with at most N coordinates. An actual global maximizer means an array attaining this full supremum, with all centered finite-third-moment laws admitted as competitors.

Theorem 1.2 (Exact structure of small-Lyapunov globals)**.** *Suppose each row is an actual global maximizer at its own finite horizon N_n, and $L_n \to 0$. Then every sufficiently late original array has identically distributed nonzero two-point coordinates and zero coordinates. The assertion applies to every original representation. Equivalently, there exists an absolute $\varepsilon_0 > 0$ such that every actual full global maximizer with $L < \varepsilon_0$ is padded IID binary.*

The threshold in Theorem 1.2 is ineffective. Zero padding increases the horizon without changing L, so the theorem is stated in terms of L. It does not specify the winning active count or binary probability. The universal independent sharp-constant problem outside the small-Lyapunov regime remains open.

The quantitative value theorem of Bell [3] concerns every array with $L \le \exp(-\exp(97))$. Its penalized counterexample selection and explicit local comparison are separate from the present classification of actual globals, whose finite ratios can be below C_{E}. We use the published asymptotic upper estimate

$$d_K(S/B, Z) \le C_{\mathrm{E}} L + 0.2538 L^{4/3} \qquad (L \le 10^{-3}) \tag{1.5}$$

from Shevtsova [10, Corollary 4.18, p. 303], rather than the quantitative companion theorem, for the near-Esseen selection. The moment and Fourier estimates extracted from that source are proved with their retained losses in Section 3. The contamination contact argument follows He and Cheng [9, Section 2.1, equations (7)–(12)], applied to one original coordinate rather than a common IID law. The changing-array common-scale and lattice arguments, and the relative conditional estimates required for exact phase collapse, are supplied in the proofs below.

For exact structure, an approximation with an absolute error tending to zero is insufficient. The remaining phase noise can be smaller than that error by an arbitrary amount. We first confine every support phase, then estimate conditional phase moments relative to the actual norm $\mathbb{E}|T|$. Here a phase means the residual modulo the common lattice gap. The integer count labels record the nearest lattice sites; the centered rounding residuals may depend on their labels within each coordinate, while the coordinate pairs remain independent. The integer endpoint competitor has a loss of the actual phase norm, and each comparison error is $o(\mathbb{E}|T|)$. This is the part of the argument that uses full globality.

2 Finite horizons and selected endpoints

We use closed and strict CDFs, $F_S(y) = \mathbb{P}(S \le y)$ and $F_S^-(y) = \mathbb{P}(S < y)$. Continuity of the normal target gives

$$d_K(S/B, Z) = \sup_y \max\{F_S(y) - \Phi(y/B),\ \Phi(y/B) - F_S^-(y)\}. \tag{2.1}$$

Reflection turns the strict negative branch into a closed positive branch. For positive discrepancy, the selected closed branch attains its maximum: it is upper semicontinuous and tends to zero in both tails.

Lemma 2.1 (Attainment at a fixed horizon)**.** *For each finite N, C_N is attained with zeros allowed. Every sequence with ratio tending to C_N, normalized to total variance one, has a subsequence converging coordinatewise in W_3 to a maximizing array.*

Proof. A symmetric binary coordinate padded with zeros gives a positive benchmark $c_0 = \Phi(1) - 1/2$. Since discrepancy is at most one, a ratio at least c_0 implies $R \le 1/c_0$. In the product weak topology the centered N-tuples with total variance one and that cubic budget form a compact set: tightness follows from the variance bound, and bounded third moments give uniform integrability of first and second moments. The cubic budget is lower semicontinuous. Convolution is weak-continuous for fixed N. CDF bracketing against the continuous normal target shows that discrepancy is weak-continuous. Finally $R \ge \sum_i (\mathbb{E}X_i^2)^{3/2} \ge N^{-1/2}$, so discrepancy divided by R is upper semicontinuous. Arrays outside this compact set have smaller ratio than the benchmark.

Along an extremizing subsequence, the discrepancies converge. Their cubic sums therefore converge to the discrepancy of the limiting array divided by C_N. Lower semicontinuity and maximality force equality with its actual cubic sum. Each individual cubic moment converges, because there are finitely many nonnegative terms. Weak convergence together with third-moment convergence is W_3 convergence. □

The same supremum is obtained if every individual variance is required to be positive, by replacing zero laws with arbitrarily small centered bounded laws; that restricted class need not attain it. A separate probability-Hessian result gives at most four support points in every original global law and some equally optimal three-point representative preserving each variance and price [2, Theorem 3.2]. The exact proof below works on the original laws directly and does not use an existential replacement to change their support.

3 Moment losses and a common scale

This section proves the approximate core used in both main theorems. We normalize $B = 1$ and omit row indices. Thus $R = L$. Put $v_i = \mathbb{E}X_i^2$, $\sigma_i = \sqrt{v_i}$, $\rho_i = \mathbb{E}|X_i|^3$, $\kappa_i = \mathbb{E}X_i^3$, and $b_i = \mathbb{E}|X_i|$. Define

$$\delta_i = \sqrt{10}\rho_i - |\kappa_i| - 3b_i v_i.$$

Lemma 3.1 (Retained losses)**.** *Under* (1.3)*,*

$$\sum_i \delta_i = o(R), \quad \sum_i \mathbb{E}[|X_i|^3 \mathbf{1}_{\{|X_i| > R^{1/6}\}}] = o(R), \quad \sum_i |\kappa_i| - \Big|\sum_i \kappa_i\Big| = o(R), \tag{3.1}$$

and $\sum_i \sigma_i^3/R \to 1/\beta_*$. After one reflection of a row, let P_i^* be the Esseen two-point law of variance v_i with the same positive orientation for every i. Then

$$\sum_i W_3^3(X_i, P_i^*) = o(R). \tag{3.2}$$

Proof. First we record the equality and stability part of the scalar moment inequality. Put $s = \sqrt{p_* q_*}$, $a = \sqrt{q_*/p_*}$, $b = \sqrt{p_*/q_*}$ and $t = t_*$. For a centered variance-one law oriented to have nonnegative third moment, set

$$Q(x) = -6sx^2 + 6tx + 6s, \qquad f(x) = x^3 - \sqrt{10}|x|^3 + 3|x|.$$

Direct factorization gives

$$Q(x) - f(x) = \begin{cases} (\sqrt{10}-1)(x-a)^2(x+2s), & x \geq 0, \\ (\sqrt{10}+1)(-x-b)^2(-x+2s), & x \leq 0. \end{cases}$$

Since $\mathbb{E}Q(X) = 0$, $\delta(X) = \mathbb{E}(Q - f) \geq 0$; equality forces the two Esseen support points, with their centered probabilities. This also checks the equality statement needed from Shevtsova [10, Theorem 3.1, pp. 256–262].

For $c_0 = 2p_*(\sqrt{10}-1) > 0$, each factor is at least c_0 times the cube of the distance to its half-line contact. Round a variance-one law to a on the positive half-line and to $-b$ on the negative half-line. If e is its cubic rounding cost, then $e \leq \delta(X)/c_0$ and centering gives $|\mathbb{P}\{\text{rounded value} = a\} - p_*| \leq e^{1/3}/(a+b)$. Couple the rounded law to the exact Esseen probabilities. The inequality $(u+v)^3 \leq 4(u^3+v^3)$ yields

$$W_3^3(X, W_+) \leq 4e + 4(a+b)^2 e^{1/3}.$$

After rescaling coordinates and applying Hölder,

$$\frac{\sum_i W_3^3(X_i, P_i^{\text{oriented}})}{R} \leq \frac{4}{c_0}\bar{\delta} + 4(a+b)^2 c_0^{-1/3}\bar{\delta}^{1/3}, \qquad \bar{\delta} = R^{-1}\sum_i \delta_i. \tag{3.3}$$

Here the orientations initially follow the signs of individual κ_i; zero laws have zero cost.

The proof on pp. 289–296 of Shevtsova [10] permits retention of its nonnegative losses. We give the details of this extraction. Put $U = R^{1/6}$, $Y_i = X_i \mathbf{1}_{\{|X_i| \leq U\}}$, $Z_i = Y_i - \mathbb{E}Y_i$, and $T_i = \mathbb{E}[|X_i|^3 \mathbf{1}_{\{|X_i| > U\}}]$. The source's I_{14}, I_{15}, I_{16} comparisons, with its truncation parameter $u = 1$, and Lemma 4.3 on pp. 272–274 give

$$\Delta \leq \phi(0)(I_{16} + R/2) + O(R^{7/6}),$$

$$I_{16} = \frac{\sqrt{10}R}{6} - \frac{1}{6}\sum_i \delta(Z_i) + I_{17} - I_{18},$$

$$I_{17} = \frac{\sqrt{10}}{6}\sum_i (\mathbb{E}|Z_i|^3 - \mathbb{E}|Y_i|^3), \qquad I_{18} = \frac{\sqrt{10}-1}{6}\sum_i T_i.$$

The second identity is the moment decomposition before discarding its deficit. At the parameter $\sqrt{10}$ the source's moment remainder is zero. All remaining terms are of order $R^{7/6}$ or smaller, uniformly over the independent row.

Let $b_{0i} = \rho_i^{1/3}/U \leq R^{1/6}$. The bound $|\mathbb{E}Y_i| \leq T_i/U^2$ and cubic Taylor expansion show

$$\mathbb{E}|Z_i|^3 - \mathbb{E}|Y_i|^3 \leq (3b_{0i}^2 + 3b_{0i}^4 + b_{0i}^6)T_i.$$

For $R \leq 1/4096$, $b_{0i} \leq 1/4$ and the bracket is less than $1/3$. Since $\sqrt{10}/3 < (\sqrt{10}-1)/2$, the displayed decomposition implies

$$\Delta \leq C_{\mathrm{E}} R - \frac{\phi(0)}{6}\sum_i \delta(Z_i) - \frac{\phi(0)(\sqrt{10}-1)}{12}\sum_i T_i + O(R^{7/6}).$$

Near-Esseen competition forces both losses to be $o(R)$. Cubic Taylor bounds give $|\rho_i - \rho(Z_i)| + |\kappa_i - \kappa(Z_i)| \leq KT_i$, while $\mathbb{E}|Z_i| \operatorname{Var} Z_i - \mathbb{E}|X_i| \operatorname{Var} X_i \leq b_{0i}^2 T_i$. Consequently $\delta_i \leq \delta(Z_i) + K'T_i$, proving the first two claims in (3.1) and the individually oriented version of (3.2).

To retain orientation we use the aggregate low-frequency integral, before the source's individual cubic triangle inequality. Write $K_3 = \sum_i \kappa_i$, $\nu = 1 + \sum_i b_i v_i/R \in [1,2]$ and $T = 2\pi/(\nu R)$. On each fixed compact frequency interval the just-proved tail estimate gives

$$\chi(u) = e^{-u^2/2}(1 - iK_3u^3/6) + o(R),$$

uniformly there. Indeed the summed Taylor remainders are at most $K[UR|u|^4 + \sum_i T_i|u|^3] = o(R)$ and the summed logarithmic quadratic errors are $O(\sum_i v_i^2 + \sum_i \rho_i^2) = o(R)$.

We use Lemma 4.1 and (4.4)–(4.7) on pp. 263–267 of the source. Its smoothing kernel is

$$K(t) = \frac{1-|t|}{2} + \frac{i}{2}\left\{(1-|t|)\cot(\pi t) + \frac{\operatorname{sign} t}{\pi}\right\}, \qquad 0 < |t| < 1.$$

The constants κ_1 and $t_1 \in (1/2, 1)$ below are the universal constants in the source's (4.4)–(4.6); in particular $t^2(1 - 4\pi\kappa_1 t)$ is strictly positive on $0 < t \le t_1$. Choose a fixed small $t_0 > 0$ so that its deleted-product bound (4.7) has Gaussian exponent at least $1/4$ for $0 \le u \le t_0 T$. The one-coordinate second-order Taylor bound against a same-variance normal gives

$$R^{-1}|\chi(u) - e^{-u^2/2}| \le K|u|^3 e^{-u^2/4} \quad (0 \le u \le t_0 T).$$

The Prawitz kernel satisfies $2|K(u/T)|/T \le 1.0253/(\pi u)$ and converges to $1/(\pi u)$. Dominated convergence therefore evaluates the source's low integral as $\phi(0)|K_3|/6 + o(R)$. Its middle interval is exponentially small: the function $t^2(1 - 4\pi\kappa_1 t)$ is positive on the chosen $[t_0, t_1]$. At the final endpoint $|K(1-u)| \le Ku$ and $1 - \cos(2\pi u) \ge 8u^2$, giving $O(T^{-2})$. The Gaussian endpoint terms are at most $\sqrt{\pi/2}/T + O(T^{-2})$. Thus

$$\Delta \le \phi(0)\left\{\frac{R}{2} + \frac{1}{2}\sum_i b_i v_i + \frac{|K_3|}{6}\right\} + o(R) = C_{\mathrm{E}}R - \frac{\phi(0)}{6}\left\{\sum_i \delta_i + \sum_i |\kappa_i| - |K_3|\right\} + o(R). \quad (3.4)$$

This proves the orientation loss. Reflect so $K_3 \ge 0$. Moment continuity under the cubic couplings gives

$$\sum_i |\kappa_i - \kappa(P_i^{\mathrm{oriented}})| \le 3\left(R + \beta_* \sum_i \sigma_i^3\right)^{2/3}\left(\sum_i W_3^3(X_i, P_i^{\mathrm{oriented}})\right)^{1/3} = o(R).$$

The negatively oriented target coordinates then have $\sum \sigma_i^3 = o(R)$; reflecting just those targets costs $o(R)$ in cubic transport. This proves the common-orientation assertion.

Finally Shevtsova [10, Theorem 4.14, pp. 300–303] gives, for each fixed $c > 2\phi(0)/3$,

$$\Delta \le cR + K(c)\sum_i \sigma_i^3 + O_c(R^{4/3}),$$

where its Bernoulli envelope is

$$K(c) = \sup_{0<p\le 1/2} \frac{\phi(0)(3 + 1 - 2p)/6 - c\{p^2 + (1-p)^2\}}{\sqrt{p(1-p)}}.$$

At $c = C_{\mathrm{E}}$ its unique maximizing parameter is p_*, $K(C_{\mathrm{E}}) = 0$, and $K'(C_{\mathrm{E}}) = -\beta_*$. The numerator's exact maximum and uniqueness follow either by differentiation or from the sharp moment equality above; parameters near zero are separated since $c > 2\phi(0)/3$. Any subsequential limit $z \in [0,1]$ of $\sum \sigma_i^3/R$ satisfies $C_{\mathrm{E}} \le c + K(c)z$ on both sides of C_{E}. Differentiation there forces $1 - \beta_* z = 0$. □

Lemma 3.2 (An actual Fourier window)**.** *For every fixed $\epsilon > 0$, a sufficiently small fixed $t_0 > 0$ gives*

$$T' = (1+\epsilon)\frac{2\pi}{\nu R}, \qquad H = 2\int_{t_0}^{1} |K(u)|\,|\chi(T'u)|\,du,$$

with $\nu \to \nu_*$ and

$$\liminf H/R \geq \frac{\phi(0)\nu_*\epsilon}{2(1+\epsilon)} > 0. \tag{3.5}$$

For one fixed such window there are $\eta > 0$ and $e_n \to 0$ such that simultaneously for every subset of coordinates A,

$$\sum_{i\in A} v_i \leq \eta^{-1}\frac{\sum_{i\in A}\rho_i}{R} + e_n. \tag{3.6}$$

Moreover $\sum_i W_2^2(X_i, P_i^*) \to 0$.

Proof. Common-orientation moment continuity from Lemma 3.1 gives $\nu \to 1 + 3/\sqrt{10} = \nu_*$ and $K_3/R \to 1/\sqrt{10}$. Repeat the source's Prawitz inequality at the enlarged cutoff T', choosing t_0 smaller so its low-frequency domination remains valid. Keep the actual high integral H rather than extending a previous high-frequency upper estimate beyond its domain. The low integral is $\phi(0)|K_3|/6 + o(R)$ and the Gaussian endpoint terms are at most $\phi(0)\nu R/[2(1+\epsilon)] + O(R^2)$. Near-Esseen competition gives (3.5).

For any centered law and an independent copy,

$$|\chi_i(u)|^2 \leq 1 - v_i u^2 + \tfrac{4}{3}\rho_i|u|^3.$$

This follows from $\cos x \leq 1 - x^2/2 + |x|^3/6$ and $\mathbb{E}|X_i - X_i'|^3 \leq 8\rho_i$. Independence gives for a subset

$$|\chi_A(u)| \leq \exp(-V_A u^2/2 + 2R_A|u|^3/3).$$

The fixed window lies in $[a/R, C/R]$ with $a > 0$. Choose $0 < \eta \leq 3/(8C)$. If $R_A/V_A \leq \eta R$, then the full product on this window is at most $\exp(-a^2 V_A/(4R^2))$. The common lower bound $H \geq hR$ implies $V_A \leq e_n = (4R^2/a^2)\log(2\sup|K|/(hR))$. Otherwise $V_A < \eta^{-1}R_A/R$. These two cases prove (3.6) with one eventual index for all subsets.

Let $E_i = W_3^3(X_i, P_i^*)$, so $\sum E_i = o(R)$. Fix $0 < \zeta < 1$. On the set $E_i > \zeta\rho_i$, the original price and, by (3.6), variance tend to zero. On its complement, Minkowski gives $\rho_i/\sigma_i^3 \leq \beta_*/(1-\zeta^{1/3})^3$ and

$$W_2^2(X_i, P_i^*) \leq E_i^{2/3} \leq v_i\left\{\frac{\zeta\beta_*}{(1-\zeta^{1/3})^3}\right\}^{2/3}.$$

On the exceptional set a centered independent same-variance coupling costs $2v_i$. Let the row tend to infinity, then let $\zeta \downarrow 0$. □

Proposition 3.3 (A variance-and-price-full Esseen core)**.** *The IID target conclusion of Theorem 1.1 holds. In gap units $h = \nu_* R$, there is a further retained set of cardinality M with*

$$MR^2 \to \beta_*^2, \quad \sup_{i\in I} W_3(\mathrm{Law}(X_i/h), \mathrm{Law}(B_{p_*} - p_*)) \to 0, \tag{3.7}$$

retaining $1 - o(1)$ of original variance and price. In these units the total original variance and price satisfy $V \sim p_* q_* M$ and $P/V \to r_*$, and its signed cubic sum divided by V tends to t_*.

Proof. Consider the variance probability measures $\mu_n = \sum_i v_i \delta_{\sigma_i/R}$. Their first moments tend to $s_0 = 1/\beta_*$, so they are tight at infinity. They have no limiting mass at zero. To prove this, discard the set $E_i > \zeta\rho_i$ as in the last lemma. On the remaining set $\rho_i/\sigma_i^3 \le K_\zeta$. For $\sigma_i \le \delta R$, its price is at most $K_\zeta \delta R$ times its variance. The subset inverse gives limiting variance at most $\eta^{-1}K_\zeta\delta$. Portmanteau on the relative-open interval $[0,\delta)$ followed by $\delta \downarrow 0$ excludes a zero atom from every limit μ. Fatou gives $\int z\,d\mu \le s_0$; first-moment uniform integrability has not yet been assumed.

At each fixed $\epsilon > 0$, (3.5) selects a frequency $u_n \in [t_0 T', T']$ with $|\chi(u_n)| \ge c_\epsilon R$. Let $w_n = Ru_n$. It lies in a compact positive interval with $\limsup w_n \le 2\pi(1+\epsilon)/\nu_*$. Independence gives $\sum_i(1 - |\chi_i(u_n)|^2) = O_\epsilon(\log(1/R))$. The same-variance binary target has defect $4p_*q_* \sin^2((a+b)\sigma_i u_n/2)$. On $\sigma_i \le KR$, the original variance-weighted defects tend to zero, and their transfer to target defects is at most

$$2|u_n|\Big(\sum_{\sigma_i \le KR} v_i^2\Big)^{1/2}\Big(\sum_i W_2^2(X_i, P_i^*)\Big)^{1/2} = o(1).$$

The untruncated target contribution is bounded by $\mu_n([K,\infty))$. First let the row tend to infinity, then $K \to \infty$. Along a weak subsequence with $w_n \to w_\epsilon$, this proves

$$\int \sin^2((a+b)zw_\epsilon/2)\,d\mu(z) = 0.$$

The limit is supported on positive integer multiples of $2\pi/((a+b)w_\epsilon)$, hence on $[s_0/(1+\epsilon),\infty)$; we used $\beta_*/(a+b) = 1/\nu_*$. Repeat for $\epsilon = 1/j$, taking further subsequences of the same weak limit. That limit is supported on $[s_0,\infty)$ and has mean at most s_0, so it is δ_{s_0}. The exact first moments converge; therefore

$$\int |z - s_0|\,d\mu_n = \int z\,d\mu_n - s_0 + 2\int (s_0 - z)_+\,d\mu_n \longrightarrow 0.$$

Trim a slowly shrinking band about s_0 and denote its count by m. It retains total variance $1-o(1)$ and target cubic resource $\beta_* \sum_I \sigma_i^3 = R(1-o(1))$. The original price has the same retention, since uniformly over subsets

$$\sum_i |\rho_i - \beta_*\sigma_i^3| \le 3\left(R + \beta_*\sum_i \sigma_i^3\right)^{2/3}\left(\sum_i E_i\right)^{1/3} = o(R).$$

Counting retained variances gives $mR^2/\beta_*^2 \to 1$. Replace the core's individually scaled targets by $W_+/\sqrt{m}$. Their summed squared scale error tends to zero, and their summed cubed scale error is $o(R)$; both follow by expanding $\sigma_i - 1/\sqrt{m} = R(\sigma_i/R - s_0) + R(s_0 - 1/(R\sqrt{m}))$. The outside-to-zero costs are exactly its variance and original price. This proves the stated IID conclusion after restoring B.

In gap units $h = \nu_* R$, the core's total cubic transport error is $o(M)$. Remove indices whose error in gap units exceeds a positive number tending slowly to zero. Their count is $o(M)$ and the preceding uniform moment continuity estimate preserves variance and price retention. This gives (3.7); its moment limits follow by uniform W_3 continuity. □

4 A first nonzero Fourier peak and integer lifts

Lemma 4.1. *At a gap asymptotic to that in Proposition 3.3, the original variables in gap units have representations*

$$Y_i = J_i + Z_i, \qquad J_i = K_i - \mathbb{E}K_i, \quad K_i \in \mathbb{Z}, \quad \mathbb{E}Z_i = 0, \quad |Z_i| \le 1.$$

The pairs are independent between coordinates. On the retained core K_i converges uniformly in W_3, after integer translations, to $\mathrm{Bern}(p_*)$. *Outside it the summed centered integer variance and cubic price are* $o(M)$. *Writing* $K = \sum K_i$ *and* $T = \sum Z_i$,

$$Q = \operatorname{Var} T = O(1), \qquad \operatorname{Cov}(K, T) = o(\sqrt{M}), \qquad A = \operatorname{Var} K \sim p_* q_* M. \tag{4.1}$$

Proof. Fix $0 < \epsilon < 1/2$ in Lemma 3.2. The frequency in gap units window stays in a compact subinterval of $(0, 4\pi)$ containing only the alias 2π, and its actual characteristic-modulus area is at least $c/\sqrt{M}$. Uniform gap-unit W_3 convergence in (3.7) gives uniform convergence of every core characteristic function and its first two derivatives on that window. Polynomial-factor differences are controlled by the L^2 coupling and the bounded Bernoulli target, so no fourth moment is required. Around 2π those factors are nonzero and their log-modulus second derivatives are at most $-c$. Away from a fixed neighborhood the retained product is exponentially small.

Let $s_0 \to 2\pi$ be the exact retained-product modulus maximum. Its product is bounded by $\exp(-cM(s - s_0)^2)$ near that point. The area lower bound therefore yields a full-product point $s_1 = s_0 + O(M^{-1/2})$ of fixed positive height. The centered omitted original sum has variance $o(M)$, so its characteristic function changes by $o(1)$ from s_1 to s_0:

$$|\chi_{\mathrm{rem}}(s_1) - \chi_{\mathrm{rem}}(s_0)| \le |s_1 - s_0| \sqrt{\operatorname{Var} Y_{\mathrm{rem}}} = o(1).$$

The full product at the exact core maximum also has fixed positive height. Rescale the original variables by $s_0/(2\pi)$.

Align each marginal characteristic argument at 2π and write $Y_i = \theta_i + K_i + E_i$ with $E_i \in [-1/2, 1/2]$, using a fixed nearest-integer tie rule. Then

$$\mathbb{E} \sin(2\pi E_i) = 0, \quad \mathbb{E} \cos(2\pi E_i) = r_i > 0, \quad \sum_i \mathbb{E} E_i^2 \le \tfrac{1}{8} \sum_i (1 - r_i) = O(1).$$

Centering gives $Z_i = E_i - \mathbb{E} E_i$ and the claimed representation. The exact core logarithmic derivative is zero, hence $\sum_I \mathbb{E}[Y_i \sin(2\pi E_i)]/r_i = 0$. On the core $r_i \to 1$ uniformly. Removing these denominators costs $o(1)$ by Cauchy and the bounded sum of defects. Replacing the sine by $2\pi E_i$ costs $O(1) + o(\sqrt{M})$: couple Y_i to its bounded binary target and use

$$\sum_I \mathbb{E}[|Y_i||E_i|^3] \le K \sum_I \mathbb{E}|E_i|^3 + \left(\sum_I \mathbb{E}|Y_i - W_i|^2 \right)^{1/2} \left(\sum_I \mathbb{E}|E_i|^6 \right)^{1/2}.$$

The omitted covariance is $o(\sqrt{M})$ by its original variance and bounded phase variance. Thus $\operatorname{Cov}(\sum Y_i, T) = o(\sqrt{M})$ and $\operatorname{Cov}(K, T) = \operatorname{Cov}(\sum Y_i, T) - Q = o(\sqrt{M})$.

Core rounding gives uniform W_3 convergence of K_i to $\mathrm{Bern}(p_*)$: outside a fixed small coupling-error event its two nearest integers are exactly the binary labels, while the remaining cubic rounding error is bounded by a constant times the cubic target-coupling error. For the omitted laws, $J_i = Y_i - Z_i$ and $|Z_i| \le 1$ imply

$$\sum_{\mathrm{rem}} \operatorname{Var} K_i \le 2 \sum_{\mathrm{rem}} \operatorname{Var} Y_i + 2Q = o(M), \quad \sum_{\mathrm{rem}} \mathbb{E}|J_i|^3 \le 4 \sum_{\mathrm{rem}} \mathbb{E}|Y_i|^3 + 4Q = o(M).$$

All prices in the last expression remain separate original-coordinate prices. These estimates also give $A \sim p_* q_* M$. □

5 Joint integer and phase estimates

We first work under the hypotheses of Lemma 4.1. Write $J_i = K_i - \mathbb{E}K_i$, $K = \sum K_i$, $\mu = \mathbb{E}K$, $T = \sum Z_i$, $A = \operatorname{Var} K$, $Q = \operatorname{Var} T$, $S = K - \mu + T$, and $V = \operatorname{Var} S$. The retained core has size M, converges uniformly in W_3 to $\mathrm{Bern}(p)$ for a fixed $0 < p < 1$, and retains all but $o(M)$ of integer variance and cubic price. In addition $Q = O(1)$ and $\operatorname{Cov}(K, T) = o(\sqrt{M})$. Therefore

$$A \sim p(1-p)M, \quad V = A + o(\sqrt{A}), \quad \gamma = \frac{\sum_i \mathbb{E}J_i^3}{A} \longrightarrow 1 - 2p. \tag{5.1}$$

Lemma 5.1 (Global integer estimates)**.** *Uniformly over every integer j, with $x_j = (j - \mu)/\sqrt{A}$,*

$$\mathbb{P}(K \le j) = \Phi(x_j) + \frac{\phi(x_j)}{\sqrt{A}} \left\{ \frac{1}{2} + \frac{\gamma}{6}(1 - x_j^2) \right\} + o(A^{-1/2}), \tag{5.2}$$

$$\sqrt{A}\mathbb{P}(K = j) = \phi(x_j) + o(1). \tag{5.3}$$

Also, uniformly over every integer k,

$$\mathbb{E}[T^2 \mathbf{1}_{\{K=k\}}] \le K_0 Q/\sqrt{A}. \tag{5.4}$$

All assertions remain uniform after a fixed number of coordinate deletions when their shifts and moments satisfy the same displayed hypotheses.

Proof. Each retained integer law has its probabilities at $0, 1$ bounded below. After at most any fixed number of deletions its characteristic product obeys

$$|\chi_{\text{deleted}}(u)| \le K \exp(-cM \sin^2(u/2)), \quad |u| \le \pi. \tag{5.5}$$

Other factors can only reduce the modulus. Uniform W_3 convergence to the bounded binary target gives uniform integrability of the core third moments and convergence of the first three characteristic derivatives on compact intervals. In a fixed small zero cell its logarithm has expansion

$$\log \chi_I(u) = -Wu^2/2 - i\kappa_I u^3/6 + \mathcal{E}(u), \quad |\mathcal{E}(u)| \le M(\epsilon_n |u|^3 + Ku^4), \quad \epsilon_n \to 0.$$

The fourth power here is a remainder for the bounded target, not a fourth-moment assumption on the actual laws. Gaussian-weighted Fourier integration divided by $|u|$ gives error $O(\epsilon_n/\sqrt{M} + 1/M)$; the complement of the zero cell is exponentially small by (5.5).

For precise CDF inversion take the Gaussian–Hermite lattice sequence

$$b_j = \frac{\phi((j - \mu_I)/\sqrt{W})}{\sqrt{W}} \left\{ 1 + \frac{\kappa_I}{6W^{3/2}} H_3((j - \mu_I)/\sqrt{W}) \right\}, \quad H_3(x) = x^3 - 3x.$$

Its mass error is exponentially small by Poisson summation; correct that mass at the integer nearest μ_I. The corrected Fourier alias numerator vanishes at zero and is $O(\mathrm{poly}(M) e^{-cM}|u|)$ there, so division by the CDF denominator $1 - e^{-iu}$ is integrable. Euler–Maclaurin summation gives the closed half-step $\phi(x)/(2\sqrt{W})$ and the integrated cubic term, with uniform $O(W^{-1})$ error. Inversion without the denominator gives the local estimate for the core.

Convolve with the omitted integer block. Its centered variance and additive cubic price are $o(A)$. Transfer the half-step and cubic correction using their bounded derivatives and its $o(\sqrt{A})$ first absolute moment. Replace only the smooth leading normal CDF coordinatewise by same-variance normals; third-order Taylor replacement costs $K \sum_{\text{rem}} \mathbb{E}|J_i|^3/A^{3/2} = o(A^{-1/2})$. This proves (5.2). Convolution and uniform continuity of ϕ prove (5.3).

For (5.4), expand the Fourier transform of $T^2\mathbf{1}_{\{K=k\}}$ without dividing by any characteristic factor. The diagonal numerators have summed magnitude at most Q; cross numerators have factors $\mathbb{E}[Z_i e^{iuJ_i}]$, whose magnitude is at most $|u|\sqrt{\mathrm{Var}(K_i)\mathbb{E}Z_i^2}$ by centering. Their squared sum is at most u^2AQ. The product after at most two extra deletions is bounded by (5.5). Integration of $KQ(1+Au^2)e^{-cAu^2}$ proves the assertion. These proofs are uniform under fixed deletions; they use products with those factors removed and no characteristic-function quotients outside the zero cell. □

Lemma 5.2 (Joint local limit)**.** *Along any subsequence with $T \Rightarrow U$,*

$$((K-\mu)/\sqrt{A}, T) \Rightarrow (G, U),$$

where G is standard normal independent of U, and $\mathbb{E}U = 0$. For every sequence $(k-\mu)/\sqrt{A} \to x \in \mathbb{R}$,

$$\sqrt{A}\mathbb{P}(K = k, T \in du) \Rightarrow \phi(x)\,\mathrm{Law}(U)(du). \tag{5.6}$$

Proof. Bounded second phase moments give tightness and first-moment uniform integrability. Fix real u, v. On the core, $g_i(v) = \mathbb{E}e^{ivZ_i} \to 1$ uniformly. At $s = u/\sqrt{A}$, expand the logarithm of $\mathbb{E}e^{isJ_i+ivZ_i}/g_i(v)$ through second order. The total third-order error is $O(M|s|^3) = o(1)$. Its quadratic coefficient divided by A tends to -1, by uniform integrability of J_i^2 and uniform $Z_i \to 0$ in L^2; the squared first coefficients contribute only $O(Q)/A$.

For the linear term use $\mathbb{E}[J_i e^{ivZ_i}] = iv\mathbb{E}[J_iZ_i] + O_v(\mathbb{E}[|J_i|Z_i^2])$. The summed error is $o(\sqrt{M})$: the bounded target contributes KQ, and the coupling remainder contributes $o(\sqrt{M})(\sum \mathbb{E}Z_i^4)^{1/2}$. Correcting the $g_i(v)$ denominators costs $K_v \sum_I (\mathbb{E}Z_i^2)^{3/2} = o(1)$. The omitted covariance is $o(\sqrt{A})$, so the core linear sum is also $o(\sqrt{A})$. We obtain

$$\chi_I(u/\sqrt{A}, v) = e^{-u^2/2+o(1)}\chi_{T_I}(v).$$

Replacing the omitted normalized integer count by zero costs at most $|u|\sqrt{\mathrm{Var}\, K_{\mathrm{rem}}/A} = o(1)$, and leaves its complete phase law. Multiplication proves the joint weak limit.

For the local assertion invert only the integer coordinate. On a fixed zero cell the twisted core factors have uniform C^2 convergence to the centered binary factor, are nonzero, and have log-modulus second derivatives at most $-c$. Their summed first derivative is at most $K_v\sqrt{A}$. Thus

$$\left|\prod_I \mathbb{E}e^{isJ_i+ivZ_i}\right| \le e^{K_v|s|\sqrt{A}-cAs^2}.$$

Outside this cell the uniformly binary-like twisted product is exponentially small. With $w = s\sqrt{A}$, dominated inversion and the just-proved joint characteristic limit give $\phi(x)\mathbb{E}e^{ivU}$. The masses converge by (5.3), and (5.4) gives tightness of these finite positive phase measures. Fourier tests followed by compact approximation establish (5.6). □

Lemma 5.3 (Endpoints and localization)**.** *If $y/\sqrt{A} \to x \in \mathbb{R}$, let $j = \lfloor y+\mu \rfloor$ and, after a subsequence, $\theta = y+\mu-j \to \theta_0 \in [0,1]$. Then*

$$\limsup \sqrt{A}\{F_S(y) - \Phi(y/\sqrt{V})\} \le \phi(x)\{\tfrac{1}{2} + \tfrac{1-2p}{6}(1-x^2) - \mathbb{E}\{\theta_0 - U\}\}, \tag{5.7}$$

$$\liminf \sqrt{A}\{F_S^-(y) - \Phi(y/\sqrt{V})\} \ge \phi(x)\{-\tfrac{1}{2} + \tfrac{1-2p}{6}(1-x^2)\}. \tag{5.8}$$

If $|y|/\sqrt{A} \to \infty$, both discrepancies multiplied by $\sqrt{A}$ tend to zero.

Proof. The actual-variance and count-variance normal baselines differ uniformly by $o(A^{-1/2})$ from (5.1). The exact closed crossing difference from the count CDF at j is

$$\sum_{\ell\in\mathbb{Z}}\Big\{\mathbb{P}(K=j+\ell,T\le\theta-\ell)-\mathbf{1}_{\{\ell\le0\}}\mathbb{P}(K=j+\ell)\Big\}.$$

For $|\ell|>H\ge3$ any crossing requires $|T|\ge|\ell|-1$. The all-layer bound therefore gives total crossing probability at most $KQ/(H\sqrt{A})$, uniformly over all real thresholds. First truncate to $|\ell|\le H$. The moving closed half-lines in (5.6) give an upper bound with $\mathbb{P}(U\le\theta_0-\ell)$. Now let $H\to\infty$. The resulting sum is

$$\phi(x)\mathbb{E}\lfloor\theta_0-U\rfloor=\phi(x)\{\theta_0-\mathbb{E}U-\mathbb{E}\{\theta_0-U\}\}.$$

Combine it with the count expansion about the real threshold, whose coefficient contains $-\theta_0+1/2$. This proves the closed bound. For strict CDFs moving open half-lines give a lower bound with $\lceil\theta_0-U\rceil-1\ge\theta_0-U-1$, proving the strict bound. These conventions include limiting phase atoms at integer boundaries.

For thresholds leaving every standardized compact set, fix H first. The finitely many crossing layers have masses $o(A^{-1/2})$ by the global local estimate; the remaining layers cost at most $KQ/(H\sqrt{A})$. The global integer Edgeworth correction also tends to zero there. Let the row tend to infinity and then $H\to\infty$. □

5.1 Modular concentration and transport

For the original near-Esseen rows take $p=p_*$. Their original price in gap units P and signed cubic κ satisfy $P/V\to r_*$ and $\kappa/V\to t_*$ by Proposition 3.3. The negative strict envelope has maximal limiting normalized value $\phi(0)(1/2-t_*/6)/r_*<C_{\mathrm{E}}$. The positive envelope has unique standardized maximum at zero. Near-extremality, the localization lemma, and (5.7) force $x=0$ and $\mathbb{E}\{\theta_0-U\}=0$, so U lies on one integer coset. Consequently there are phases ϑ_n with

$$\mathbb{E}\,\mathrm{dist}(S-\vartheta_n,\mathbb{Z})\longrightarrow0. \tag{5.9}$$

Every subsequence admits this argument; minimizing the bounded continuous circle-distance expectation over the compact circle gives phases for the full sequence.

We give the transport implication directly. For $q\in(0,1)$, let $t=2q-1$ and define the continuous periodic triangle

$$P_q(u)=\begin{cases}-2(1-q)u, & 0\le u\le q,\\ 2q(u-1), & q\le u\le1.\end{cases}$$

Its mean is $-q(1-q)$ and its nonzero Fourier coefficients are $(1-e^{-2\pi ikq})/(2\pi^2k^2)$. Zero bias gives $\mathbb{E}S^*=\kappa/(2V)$ and $|\mathbb{E}e^{iuS^*}|\le1/(|u|\sqrt{V})$, by its defining identity. The test $-tx+P_q(x-\vartheta)$ is 1-Lipschitz. The Wasserstein dual and its absolutely convergent Fourier series yield

$$W_1(S,S^*)\ge\frac{t\kappa}{2V}+q(1-q)-2\mathbb{E}\,\mathrm{dist}(S-\vartheta,\mathbb{Z})-\frac{\zeta(3)}{\pi^3\sqrt{V}}. \tag{5.10}$$

With $q=q_*$ and (5.9), the lower limit is $t_*^2/2+p_*q_*=r_*/2$. The original additive bound $W_1(S,S^*)\le P/(2V)\to r_*/2$ matches it, proving $\mathcal{B}(X_n)\to1$.

Modular concentration also strengthens the actual rounding. It gives $|\mathbb{E}e^{2\pi iS}|\to1$, hence independence implies $\sum_i(1-r_i^2)\le-\log\prod_i r_i^2\to0$. The argument-aligned errors of Lemma 4.1 therefore have $Q=\sum_i\mathbb{E}Z_i^2\to0$. Since $|Z_i|\le1$, their actual independent couplings give

$$\sum_i W_2^2(Y_i,J_i)\le Q\to0,\qquad\sum_i W_3^3(Y_i,J_i)\le Q\to0.$$

Restore the gap units to obtain (1.4). This completes Theorem 1.1.

Remark 5.4. The finite inequality $C \le C_{\mathrm{E}}\mathcal{B}(X)$ is false. To see why, replace the part of the standard normal law in $[-a,a]$ by atoms at $-a,0,a$, preserving its mass and second moment. If $P = \mathbb{P}(|Z| \le a)$ and $r = \mathbb{E}[Z^2\mathbf{1}_{\{|Z|\le a\}}]/(a^2P)$, these masses are $Pr/2, P(1-r), Pr/2$, with $0 < r < 1/3$. The discrepancy is $P(1-r)/2$. Its zero-biased law instead replaces the same interval by a uniform law, and $W_1 = (aP/2)((1-r)^2 + r^2)$. The ratio of the Berry–Esseen statistic to $\mathcal{B}$ is $(1-r)/(2a((1-r)^2+r^2)) \ge 1/(2a)$, which diverges as $a \downarrow 0$. Both statistics tend to zero. Theorem 1.1 uses competitive near-Esseen value as well as small L.

6 Full contacts and every support phase

For the remainder of the proof each row is an actual full global maximizer and $L \to 0$. The active count is at least L^{-2}, by Lyapunov and the power-mean inequality, so its horizon tends to infinity. An IID Esseen array at the same horizon has ratio tending to C_{E} [6, pp. 160–163]; globality and (1.5) therefore give $C \to C_{\mathrm{E}}$. Theorem 1.1 and all its preceding estimates apply.

Lemma 6.1 (Original support coercivity)**.** *Normalize total variance to one and let $R = L$. For all sufficiently late global rows every original support point satisfies $|u| \le 246R^{1/3}$. At a positive closed maximizing threshold z, put $D = F(z) - \Phi(z)$, $\alpha = \phi(z)$, $A_0 = (3D + z\alpha)/2$. For coordinate i, write $v = \mathbb{E}X_i^2$, $\rho = \mathbb{E}|X_i|^3$ and $M_i = \mathbb{E}[X_i|X_i|]$. The full contact inequality is*

$$F_{-i}(z-u) + (\alpha + 3CM_i)u + A_0u^2 - C|u|^3 \le F(z) + A_0v - C\rho, \tag{6.1}$$

with equality at every topological support point of X_i.

Proof. We use the independent-coordinate form of the contamination derivation of He and Cheng [9, Section 2.1, equations (7)–(12)]. Their variation changes the common IID law; here it changes just one original coordinate and retains its contribution to the additive absolute cubic price. Replace the unstandardized coordinate law by $(1-e)\,\mathrm{Law}(X_i) + e\delta_u$, keep its unstandardized closed threshold, and recenter and restandardize. The derivatives of mean, total variance, and original absolute cubic price are respectively $u, u^2 - v, |u|^3 - \rho - 3uM_i$. The ratio derivative gives (6.1) for every real u. Its integral under X_i is equality. The deleted closed CDF is upper semicontinuous, so equality extends from a full-measure set to every topological support point, including arbitrarily small atoms.

Eventually $C \ge 199/500$, $R < 1/750$ and the general bound of Chen and Shao [5, Theorem 2.1] gives $C \le 41/10$. Lyapunov gives $v^{3/2} \le \rho \le R$ and $v < 1/2$. The deleted normal approximation has uniform error at most $(123/10)R$. Put $s = \sqrt{1-v}$. Uniform Gaussian scale estimates are

$$|\Phi(z/s) - \Phi(z) - z\phi(z)v/2| \le 2v^2,$$

$$|\phi(z/s)/s - \phi(z)| \le 3v/5, \qquad |z\phi(z/s)/(2s^3) - z\phi(z)/2| \le 4v.$$

They follow by differentiating in v and using $\|(1-x^2)\phi(x)\|_\infty = \phi(0) < 2/5$ and $|x(3-x^2)\phi(x)| < 4$. The cubic translation remainder is at most $\phi(0)|u|^3/(6s^3) < |u|^3/5$. Substitute the deleted approximation into contact equality and use $|M_i| \le v$, $D \le (41/10)R$, $v^2 \le \rho \le R$. This gives

$$(C - 1/5)|u|^3 \le \frac{1023}{40}R + \frac{129}{10}v|u| + \left(\frac{123}{20}R + 4v\right)u^2.$$

For $|u| = R^{1/3}w$, divide by R and use $v \le R^{2/3}$:

$$\frac{99}{500}w^3 \le \frac{1023}{40} + \frac{129}{10}w + \frac{203}{20}w^2.$$

For $w \geq 1$ the right side is at most $(389/8)w^2$, so $w \leq 48625/198 < 246$. The other case is immediate. Reflection of a strict negative threshold gives the same bound. □

In the selected gap units write the original laws as $Y_i = J_i + Z_i$ from Lemma 4.1. Put

$$V = \sum_i \mathbb{E}Y_i^2, \quad d = V^{-1/2}, \quad P = \sum_i \mathbb{E}|Y_i|^3, \quad Q = \sum_i \mathbb{E}Z_i^2, \quad T = \sum_i Z_i,$$

and select a positive closed threshold y, with $z = y/\sqrt{V}$. The preceding results give

$$P/V \to r_*, \quad \kappa/V \to t_*, \quad Q \to 0, \quad z \to 0, \quad D/d \to \phi(0)(1/2 + t_*/6), \quad \mathrm{Cov}(K, T) = o(\sqrt{V}). \tag{6.2}$$

The support bound is $|Y_i| \leq H = O(V^{1/3})$. The uniform retained core and resource retention give $\sup_i \mathbb{E}|Y_i|^3 = o(V)$. Consequently $\max_i \mathbb{E}Y_i^2 = o(V)$ and $\sum_i \mathbb{E}Y_i^4 = O(HV)$.

Lemma 6.2 (Whole-support fractional confinement)**.** *For the actual lifts just constructed,*

$$\eta_n := \max_i \|Z_i\|_\infty \longrightarrow 0. \tag{6.3}$$

The nearest representative of $\{y + \mu\}$ on the circle tends to zero. If $t = \mathbb{E}|T| > 0$, then $Q = o(t)$.

Proof. The modular concentration and $Q \to 0$ already proved imply that $\{y + \mu\}$ tends to zero on the circle. This also follows directly from the saturated closed envelope and $T \to 0$ in L^2.

Delete any one coordinate. Let its original variance be v_i, its deleted original variance be $w_i = V - v_i$, and its integer deleted variance be A_i. Uniformly in the deletion,

$$w_i \sim V, \quad A_i - w_i = o(\sqrt{V}), \quad \kappa(K_{-i})/A_i - \kappa(Y_{-i})/w_i = o(1).$$

The total covariance is $o(\sqrt{V})$ and each individual deleted covariance is at most $(H+1)\sqrt{Q} = o(\sqrt{V})$. Cubic differences are at most $K(\sqrt{HVQ} + HQ + Q) = o(V)$. The retained core still damps all deleted products. Lemma 5.1 therefore applies uniformly after these deletions. Its fourth-moment version has a zero-cell CDF remainder $O(H/V) = o(d)$ and exponentially small outer-frequency error; it follows by the same inversion with log remainder $O(HVu^4)$.

For $|u| \leq H$, set

$$x_i = (y - u)/\sqrt{w_i}, \quad \gamma_i = \kappa(Y_{-i})/w_i, \quad G_i(u) = \Phi(x_i) + w_i^{-1/2}\phi(x_i)\{1/2 + \gamma_i(1 - x_i^2)/6\},$$

and $\tau_i(u) = \{y - u + \mu_{-i}\}$. If $\tau_i(u)$ stays in $[\epsilon, 1 - \epsilon]$, the deleted all-layer bound gives a crossing error at most

$$\sum_{\ell \in \mathbb{Z}} \frac{KQd}{(\ell - \tau_i(u))^2} = o(d).$$

No stationary covariance is needed after deletion: the quotient-free second-moment integrand is bounded by $K[Q + A_i Q u^2]e^{-cVu^2}$. Thus uniformly over this phase range,

$$F_{-i}(y - u) = G_i(u) - d\phi(0)\tau_i(u) + o(d). \tag{6.4}$$

In gap units the contact score and level are

$$g_i(u) = F_{-i}(y - u) + b_i u + A_0 u^2 - \lambda|u|^3, \quad \ell_i = F(y) + A_0 v_i - \lambda\rho_i,$$

$$b_i = d\phi(z) + 3\lambda\mathbb{E}[Y_i|Y_i|], \quad \lambda = D/P = Cd^3, \quad A_0 = (3D + z\phi(z))/(2V).$$

They satisfy $g_i \le \ell_i$, with equality on all original supports. We claim

$$G_i(u) + b_i u + A_0 u^2 - \lambda|u|^3 - \ell_i \le o(d) \tag{6.5}$$

uniformly over i and $|u| \le H$. At $u = 0$ the leading deleted-variance term is $z\phi(z)v_i/(2V)$, exactly the same as its term in $A_0 v_i$. The second-order error is $O(d^4 v_i^2) \le O(d^4 H^4) = o(d)$; the remaining level terms are $O(d^3 v_i + d^3\rho_i) = o(d)$. For changes from zero to u, cancel the normal quadratic threshold term before expansion. The linear derivative error is $O(d^3 v_i)$, the remaining quadratic terms are $O(d^3 u^2)$, and the normal cubic remainder is at most $(\phi(0)/6 + o(1))d^3|u|^3$. The even Edgeworth correction has changes at most $K(d^2|z||u| + d^3u^2)$. Thus, since $C \to C_{\mathrm{E}} > \phi(0)/6$, the left side of (6.5) is at most

$$-cd^3|u|^3 + Kd^3u^2 + Kd^3 v_i|u| + Kd^2|z||u| + o(d).$$

Young's inequality absorbs the positive terms, leaving $O(d^3 + d^3 v_i^{3/2} + d^{3/2}|z|^{3/2}) = o(d)$, since $v_i^{3/2} \le \rho_i = o(V)$. This proves the claim.

If an original support phase $u + \mathbb{E}K_i$ stayed a positive circle distance from zero, its deleted phase $\tau_i(u)$ would stay away from the boundary because $\{y + \mu\} \to 0$. Equations (6.4) and (6.5) would give $g_i(u) < \ell_i$, contradicting its exact original contact. Hence every support phase approaches its integer coset uniformly. The aligned errors lie in $[-1/2, 1/2]$ and their means tend uniformly to zero, proving (6.3). Finally independence gives $\mathbb{E}T^4 \le 3Q^2 + \eta_n^2 Q$; interpolation implies $t \ge Q/\sqrt{3Q + \eta_n^2}$, hence $Q = o(t)$ when $t > 0$. □

7 Relative conditional phase moments

We now retain the actual phase norm in the local estimates. Before conditioning, we move the same integer labels to a stationary peak of the full characteristic product. This supplies the stronger covariance estimate required below.

Lemma 7.1 (The same lifts at a full peak)**.** *After a common scaling tending to one, the original laws and the same integer labels satisfy*

$$A = \operatorname{Var} K \to \infty, \quad |J_i| \le H = O(A^{1/3}), \quad \sum_i \mathbb{E}|J_i|^3 = O(A), \quad Q \to 0, \quad |\operatorname{Cov}(K, T)| \le KHQ, \tag{7.1}$$

as well as (6.3)*. The retained core still has at least cA laws with masses at two adjacent integers bounded below by $c > 0$; the selected threshold has circle phase tending to zero.*

Proof. For bounded x the integer sum satisfies $|\mathbb{E}e^{i(2\pi + x/\sqrt{A})\sum J_i}| \to e^{-x^2/2}$ uniformly on compact x intervals, by its cubic $O(A)$ log expansion. The original and integer characteristic functions differ there by at most $K\mathbb{E}|T| \to 0$. The actual full modulus at 2π tends to one, while its values at $2\pi \pm 1/\sqrt{A}$ tend to $e^{-1/2}$. Its maximum on that interval is interior, at $s = 2\pi + x_n/\sqrt{A}$ with $x_n \to 0$. Put $a = s/(2\pi) = 1 + \delta$ and keep exactly the same J_i:

$$Y_i^{\mathrm{new}} = aY_i, \qquad Z_i^{\mathrm{new}} = Z_i + \delta Y_i.$$

Then $\delta\sqrt{A} \to 0$, $\eta_{\mathrm{new}} \le \eta + |\delta|H \to 0$, and $Q_{\mathrm{new}} \le 2Q + 2\delta^2 V \to 0$. The selected threshold's phase changes by $\delta y = o(1)$.

Align the individual arguments by $\beta_i = (2\pi)^{-1} \arg \mathbb{E} e^{2\pi i Z_i^{\mathrm{new}}}$. The errors $E_i = Z_i^{\mathrm{new}} - \beta_i$ remain inside $[-1/2, 1/2]$; their nearest integer labels are unchanged. Centering them returns Z_i^{new} exactly. The sine identity and $\mathbb{E} Z_i^{\mathrm{new}} = 0$ give

$$|\beta_i| \le K\mathbb{E}|Z_i^{\mathrm{new}}|^3 \le K\eta_{\mathrm{new}}\mathbb{E}(Z_i^{\mathrm{new}})^2, \qquad \sum_i \mathbb{E}E_i^2 = Q_{\mathrm{new}}(1 + o(1)).$$

Full log-modulus stationarity at 2π is $\sum_i \mathbb{E}[Y_i^{\mathrm{new}} \sin(2\pi E_i)]/r_i = 0$. Removing denominators and replacing sine by its linear term costs at most $KH \sum_i \mathbb{E}E_i^2$, using bounded original support and $1 - r_i \le K\mathbb{E}E_i^2$. Thus $|\operatorname{Cov}(\sum Y_i^{\mathrm{new}}, T_{\mathrm{new}})| \le KHQ_{\mathrm{new}}$. Subtract Q_{new} to obtain (7.1). All resource limits are invariant under the common scaling. No stationarity has been imposed on the rounded comparator. □

Lemma 7.2 (A multiplier first-moment bound)**.** *For independent centered Z_i and $t = \mathbb{E}|\sum Z_i|$, any same-coordinate multipliers W_i with $\|W_i\|_\infty \le M_W$ satisfy*

$$\left|\sum_i \mathbb{E}W_i Z_i\right| \le 8t \left(\sum_i \mathbb{E}W_i^2\right)^{1/2} + 5M_W t. \tag{7.2}$$

Proof. Let Z_i' be independent copies and $D_i = Z_i - Z_i'$. By the second/fourth-moment Khintchine estimate, $\mathbb{E}(\sum D_i^2)^{1/2} \le 2\sqrt{3}\, t =: C_0 t$. Truncate at $b = 4C_0 t$. Markov gives probability at least $3/4$ that all other differences are small. Disjoint events with exactly one large difference therefore give $\sum \mathbb{E}[|D_i| \mathbf{1}_{\{|D_i| > b\}}] \le (4/3) C_0 t < 5t$. For $U = \sum \mathbb{E}[D_i^2 \mathbf{1}_{\{|D_i| \le b\}}]$, independence and second/fourth interpolation give $U/\sqrt{U + b^2} \le \mathbb{E}(\sum D_i^2 \mathbf{1}_{\{|D_i| \le b\}})^{1/2} \le C_0 t$, hence $U < 60t^2$. Since $\mathbb{E}W_i Z_i' = 0$, apply Cauchy to the small differences and the uniform multiplier bound to the others. If $t = 0$, independence forces every centered phase to be zero and the result is immediate. □

The inputs for the next lemma are the independent coordinate pairs (K_i, Z_i) from Lemma 7.1, with $K_i \in \mathbb{Z}$, $J_i = K_i - \mathbb{E}K_i$, and $\mathbb{E}Z_i = 0$. They obey (7.1), $\max_i \|Z_i\|_\infty \to 0$, and have a core of at least cA laws with two adjacent integer masses bounded below. This core supplies the uniform local estimates after finitely many deletions. The integer label and residual within one pair may be dependent. There is no lower bound on the positive norm t.

Lemma 7.3 (Relative conditional first moments)**.** *Under these hypotheses, whenever $t = \mathbb{E}|T| > 0$, for each fixed B_0 and uniformly over $|k - \mu| \le B_0\sqrt{A}$,*

$$c\, t \le \mathbb{E}(|T| \mid K = k) \le Kt, \qquad |\mathbb{E}(T \mid K = k)| = o(t). \tag{7.3}$$

The constants and the relative errors are uniform even if t tends to zero arbitrarily fast.

Proof. Take two independent row copies, with differences $D_i = Z_i - Z_i'$ and $D = \sum D_i$. Then $t \le \mathbb{E}|D| \le 2t$ and $\mathbb{E}(\sum D_i^2)^{1/2} \le 2\sqrt{3}\, t$. Fix a large constant B, truncate at $b = Bt$, and put

$$d_i = D_i \mathbf{1}_{\{|D_i| \le b\}}, \quad W = \sum_i d_i, \quad U = \sum_i \mathbb{E}d_i^2, \quad \Delta = \sum_i \mathbb{E}[|D_i| \mathbf{1}_{\{|D_i| > b\}}].$$

The d_i are independent, centered, and antisymmetric under exchange of copies. The preceding truncation calculation gives

$$\Delta \le Kt, \quad \sum_i \mathbb{P}(|D_i| > b) \le K/B, \quad U \le K_B t^2, \qquad K_B = O(B). \tag{7.4}$$

In the last bound use $\mathbb{E}S_b^2 \le U^2 + b^2U$ and $\mathbb{E}\sqrt{S_b} \le 2\sqrt{3}\,t$, where $S_b = \sum d_i^2$.

We establish, for bounded standardized count positions, the relative two-count estimates

$$\mathbb{E}(W^2 \mid K = k_1, K' = k_2) = U + o(t^2), \qquad \mathbb{E}(W^4 \mid K = k_1, K' = k_2) \le C_B t^4. \tag{7.5}$$

Here C_B is distinct from the variance constant K_B: we may take $C_B = O(B^3)$ for each fixed count range. Indeed its leading terms are $U^2 + b^2U$, and $U \le K_B t^2$ with $K_B = O(B)$ and $b = Bt$. We fix B before letting the row index tend to infinity. The estimates hold uniformly after one pair coordinate is specified and removed; its integer shifts have size $O(H) = o(\sqrt{A})$.

Here are the Fourier estimates retaining their relative factors. For $s = (s_1, s_2)$ in $\|s\| \le c/H$, let $f_i(u) = \mathbb{E}e^{iuJ_i}$, $\Psi(s) = \prod_i f_i(s_1) f_i(s_2)$, and

$$m_{ir}(s) = \frac{\mathbb{E}[d_i^r e^{i(s_1 J_i + s_2 J_i')}]}{f_i(s_1) f_i(s_2)}, \quad 1 \le r \le 4.$$

The denominators are bounded away from zero in this cell only. Antisymmetry gives $\mathbb{E}d_i = 0$ and $c_i = \mathbb{E}J_i d_i = -\mathbb{E}J_i' d_i$. For the untruncated differences the summed covariance is $\mathrm{Cov}(K, T) = O(HQ)$; since $|Z_i| \le 1$, interpolation gives $Q \le 2t$ eventually. Removing tails costs at most $2H\Delta$ and removing one pair at most $4Ht$, so $|\sum c_i| \le KHt$ uniformly. Lemma 7.2, applied to the independent d_i, gives

$$\sum_i \mathbb{E}[(J_i^2 + J_i'^2)|d_i|] \le K(\sqrt{HA} + H^2)t,$$

$$\sum_i a_i |c_i| \le K(H^2\sqrt{A} + H^3)t, \qquad a_i = \mathbb{E}J_i^2.$$

For these estimates the multipliers are respectively $(J_i^2 + J_i'^2)\operatorname{sign} d_i$ and $a_i|J_i|\operatorname{sign} d_i$; their summed squared moments are $O(HA)$ and $O(H^4A)$, and their uniform bounds are $O(H^2), O(H^3)$. Also $\mathbb{E}|W| \le \mathbb{E}|D| + \Delta = O(t)$. Numerator and denominator Taylor expansions now give

$$\left|\sum_i m_{i1}\right| \le Kt\left\{H\|s\| + (\sqrt{HA} + H^2)\|s\|^2 + (H^2\sqrt{A} + H^3)\|s\|^3\right\}. \tag{7.6}$$

All terms are $o(t)$ on the Gaussian integration scale. Further,

$$\left|\sum_i m_{i2} - U\right| \le KU(H\|s\| + H^2\|s\|^2), \quad \sum_i |m_{i1}|^2 \le KH^2U\|s\|^2,$$

$$\sum_i |m_{i2}| \le KU, \qquad \sum_i |m_{i3}| \le KbU, \qquad \sum_i |m_{i4}| \le Kb^2U.$$

The exact second transform is $\Psi[(\sum m_{i1})^2 + \sum(m_{i2} - m_{i1}^2)]$. Its leading term $U\Psi$ integrates to $U\mathbb{P}(K = k_1)\mathbb{P}(K' = k_2)$ exactly, and its remaining Gaussian integral is $o(t^2/A)$. For the fourth transform use

$$\Psi(M_1^4 + 6M_1^2C_2 + 3C_2^2 + 4M_1C_3 + C_4),$$

where $M_1 = \sum m_{i1}$ and C_r are the sums of the individual cumulant polynomials:

$$C_2 = \sum(m_{i2} - m_{i1}^2), \quad C_3 = \sum(m_{i3} - 3m_{i2}m_{i1} + 2m_{i1}^3),$$

$$C_4 = \sum(m_{i4} - 4m_{i3}m_{i1} - 3m_{i2}^2 + 12m_{i2}m_{i1}^2 - 6m_{i1}^4).$$

The displayed bounds give integral at most $C_B t^4/A$. Outside the zero cell use quotient-free partition formulas and delete at most four pair coordinates. Every singleton keeps a relative factor because antisymmetry gives

$$|\mathbb{E}[d_i e^{i(s_1 J_i + s_2 J_i')}]| \le K\|s\|\mathbb{E}[|d_i||J_i - J_i'|], \qquad \sum_i \mathbb{E}[|d_i||J_i - J_i'|] \le K\sqrt{A}\, t.$$

The last bound is again Lemma 7.2. Other numerators sum to at most U, bU, b^2U. Torus damping leaves a fixed polynomial in A times t^2 or t^4 times e^{-cA/H^2}, which is negligible as $H = O(A^{1/3})$. The ordinary integer local limit gives $\mathbb{P}(K = k_1)\mathbb{P}(K' = k_2) \ge c(B_0)/A$ and deletion ratios $1 + o(1)$ uniform over shifts $O(H)$. Divide these relative integrals to obtain (7.5).

If $\Delta \ge \epsilon t$, condition on an actually large coordinate and its two integer labels. Fixed-deletion ratios preserve its tail integral up to $1 + o(1)$. Other large coordinates have conditional union probability $O(1/B)$, and the truncated remainder has conditional first absolute moment at most $K\sqrt{B}\, t$. On the disjoint events with exactly one large coordinate,

$$\mathbb{E}[|D_i + \sum_{j \ne i} d_j|\mathbf{1}_{\{\text{others small}\}} \mid \text{labels and counts}] \ge (1 - K/B)|D_i| - K\sqrt{B}\, t \ge |D_i|/2$$

for sufficiently large fixed B. Thus conditional $\mathbb{E}|D| \ge c\Delta$. Otherwise $\Delta < \epsilon t$ and $\mathbb{E}|W| \ge (1-\epsilon)t$, so $U \ge (1-\epsilon)^2 t^2$. Interpolation in (7.5) gives conditional $\mathbb{E}|W| \ge c_B t$. Conditional discarded-tail first moments are at most 2Δ, by one-pair deletion ratios. Choose $\epsilon < c_B/4$ after B, to obtain conditional $\mathbb{E}|D| \ge c_B t/2$. The same decomposition gives the upper bound $K(B)t$. With $k_1 = k_2 = k$, the two rows are conditionally independent and $\mathbb{E}(|D| \mid K = K' = k) \le 2\mathbb{E}(|T| \mid K = k)$, proving the lower bound.

For the one-count mean, apply Lemma 7.2 directly to Z_i. Its zero-cell singleton sum has the analogue of (7.6), using $|\operatorname{Cov}(K, T)| = O(HQ) = O(Ht)$. Gaussian inversion gives $o(t/\sqrt{A})$, and quotient-free outer inversion retains t by the singleton multiplier bound. Divide by the central mass to obtain mean $o(t)$. Finally conditional Jensen gives

$$\mathbb{E}(|T| \mid K = k) \le \mathbb{E}(|T - T'| \mid K = K' = k) + |\mathbb{E}(T \mid K = k)| \le K(B)t.$$

□

8 An exact integer endpoint comparison

Under the preceding hypotheses, $\eta \to 0$, $Q = o(t)$ when $t > 0$, and the nearest selected integer phase θ tends to zero. For its central layer k, put

$$w(\theta) = \mathbb{P}(T > \theta \mid K = k), \qquad f(\theta) = \theta + w(\theta).$$

The strict inequality in w is dictated by the closed original CDF.

Lemma 8.1 (Relative closed loss)**.** *If $t > 0$, then at the selected layer and phase*

$$f(\theta) \ge ct, \qquad |\theta| + w(\theta) + t \le K f(\theta). \tag{8.1}$$

Proof. Condition on $K = k$ and truncate T to $X = T\mathbf{1}_{\{|T| \le 1/4\}}$. The all-layer second-moment bound, divided by the central count mass, gives $\mathbb{E}(T^2 \mid K = k) \le KQ$. Its CDF, mean and

first absolute moment change under this truncation by $O(Q) = o(t)$. Let $m = \mathbb{E}X = o(t)$ and $s = \mathbb{E}|X| \geq ct$. For $0 \leq \theta \leq 1/4$, with $u = \mathbb{P}(X > \theta)$,

$$s = 2\mathbb{E}X_+ - m \leq 2\theta + u/2 - m \leq 2(\theta + u) - m.$$

For $\theta = -b \in [-1/4, 0)$,

$$m \leq (1/4 + b)u - b \leq u/2 - b, \qquad s \leq u/2 - m.$$

Thus $\theta + u \geq s + 2m$ in the negative case, and $u \leq 2(\theta + u - m)$, $b \leq \theta + u - 2m$. These inequalities bound $|\theta| + u + t$ by a fixed multiple of $\theta + u$. Transfer the $o(t)$ truncation errors back to T. □

Proposition 8.2 (Collapse of original phase noise)**.** *Every sufficiently late actual full global row has $T = 0$ almost surely. Each original coordinate is then exactly on its constructed lattice coset.*

Proof. Suppose $t > 0$ along a subsequence. Retain exactly the same integer labels and replace the original laws $Y_i = J_i + Z_i$ by the centered integer laws J_i at the same horizon. Write k for the integer nearest $y + \mu$ and $\theta = y + \mu - k$. The exact closed-CDF difference is

$$F_Y(k + \theta - \mu) = F_K(k) - p_k w(\theta) + E_{\text{cross}}, \qquad |E_{\text{cross}}| \leq K dQ. \tag{8.2}$$

The central term is exact. Every other count layer is at distance at least $1/2$ from this threshold. Its crossing probability is bounded by the all-layer second moment divided by its squared distance; summing the inverse-square series proves the stated error.

Let $A = \operatorname{Var} K$, $V = \operatorname{Var} \sum Y_i$, $P_Y = \sum \mathbb{E}|Y_i|^3$, and $P_J = \sum \mathbb{E}|J_i|^3$. Full-peak covariance gives $|V - A| \leq KHQ$. The integer local limit gives $p_k = d\phi(z) + o(d)$, uniformly at this bounded standardized threshold. The normal threshold shift is $d\phi(z)\theta + O(d^2\theta^2)$; changing its variance costs $O(HQ/A) = o(df)$. In (8.2), $dQ = o(df)$ and both $|\theta|$ and w are controlled by f from Lemma 8.1. Consequently the selected discrepancies satisfy

$$D_Y = D_J - d\phi(z) f(\theta) + o(df(\theta)). \tag{8.3}$$

The local-mass error multiplies w and the normal error multiplies θ, so they remain relative to this actual loss.

For original cubic price use the centered path $Y_i(s) = J_i + sZ_i$, $0 \leq s \leq 1$. Its support is $O(H)$, its summed cubic moments are $O(A)$, and its fourth moments sum to $O(HA)$. Lemma 7.2, with $W_i = 3Y_i(s)|Y_i(s)|$, gives uniformly along the path

$$\left| \frac{d}{ds} \sum_i \mathbb{E}|Y_i(s)|^3 \right| \leq K(\sqrt{HA} + H^2)t = o(At).$$

Thus $|P_Y - P_J| = o(At)$, retaining the original absolute price. Both prices are asymptotic to $r_* A$. The prefactors $V^{3/2}/P_Y$ and $A^{3/2}/P_J$ differ by relative $o(t)$. At the selected competitive threshold $D_Y = O(d)$, and (8.3) gives $D_J = O(d)$ as well. Multiplying that exact comparison by the prefactors yields a positive ratio gain at least $cf \geq ct$, for all sufficiently late rows. The full objective of the integer competitor is at least its selected closed branch. It therefore contradicts full globality.

Hence $t = 0$ eventually. Independence and centeredness imply $\operatorname{Var} T = \sum_i \operatorname{Var} Z_i = 0$, so every original $Z_i = 0$ almost surely. No collision set has been held fixed during the finite comparison. □

9 Exact lattice contacts and binary support

The original laws now lie exactly on individual cosets of the same unit lattice in the selected gap units. The support coercivity bound makes them finite, while $P/V \to r_*$, $\kappa/V \to t_*$ and $z \to 0$. The retained core has two adjacent atoms with masses bounded below, so all deleted characteristic products satisfy torus damping.

Lemma 9.1 (Adjacent original contacts)**.** *Eventually every original law is binary at adjacent integer sites or zero.*

Proof. Put $v_i = \mathbb{E}Y_i^2$, $w_i = V - v_i$, and let $c_i(x)$ be the mass function of the deleted original sum. For x on its lattice coset, Fourier inversion with multiplier $e^{-iu} - 1$ gives uniformly in the deletion and in x

$$c_i(x+1) - c_i(x) = d_i^2\phi'(d_i x) + \frac{d_i^3}{2}\phi''(d_i x) + d_i^3\frac{\kappa_i}{6w_i}(H_3\phi)'(d_i x) + o(d^3), \tag{9.1}$$

where $d_i = w_i^{-1/2}$ and κ_i is the deleted signed cubic sum. In the cell $|u| \le c/H$ the deleted log characteristic function has remainder $O(HPu^4)$, since $\sum \mathbb{E}Y_j^4 \le HP = O(HV)$; summed squared variances are absorbed by this fourth-moment bound. The first-difference multiplier adds a factor $O(d|s|)$ on $u = d_i s$, so Gaussian integration costs $O(Hd^4 + d^4) = o(d^3)$. Outside this cell torus damping is at most e^{-cV/H^2}. This proves (9.1) without dividing by an atom mass.

At the selected positive closed threshold, the finite sum attains its discrepancy at an atom. Thus $y - u$ lies on the deleted coset for every test site u of the original coordinate coset. For $|u| \le H + 1$, Taylor expansion in $d_i u$, retaining z before expansion, gives

$$c_i(y-u+1) - c_i(y-u) = -d^2 z\phi(z) + d^3\phi(0)\{u - (1+t_*)/2\} + o(d^3(1+|u|)) + O(d^4|z|v_i).$$

The last term records the deletion variance; no rate for the unstandardized selected threshold is imposed.

For the centered second difference $\Delta^2 g(u) = g(u+1) - 2g(u) + g(u-1)$, the contact score has exact deleted-CDF contribution equal to the mass difference just displayed. Its term $2A_0 = d^2 z\phi(z) + 3d^2 D$ cancels the leading normal term exactly. Hence

$$d^{-3}\Delta^2 g_i(u) = \phi(0)(1+u) - C_{\mathrm{E}}\{|u+1|^3 + |u-1|^3 - 2|u|^3\} + o(1+|u|) + O(d|z|v_i). \tag{9.2}$$

The leading expression is at most $-c(1+|u|)$. For $|u| \ge 1$ use the exact absolute-cubic difference $6|u|$; for $0 \le u \le 1$ use its bound $2 + 4u^2$, giving a negative margin since $C_{\mathrm{E}} > \phi(0)$; the negative interval has still larger margin. In particular there are fixed $c, K > 0$ such that

$$\Delta^2 g_i(u) \le d^3\{KE_i - c|u|\}, \qquad E_i = 1 + d|z|v_i.$$

Two maximizing contacts at integer distance $L \ge 2$ have all intermediate grid tests bounded above by their common contact value. For $j = \lfloor L/2 \rfloor$ the discrete Green formula is

$$g_i(u_0 + j) - g_i(u_0) = -\sum_{k=1}^{L-1} G_{jk}\Delta^2 g_i(u_0 + k), \quad G_{jk} = \frac{\min(j,k)(L - \max(j,k))}{L}.$$

Here $\sum G_{jk} = j(L-j)/2$. For $L \ge 16$ the weighted average of $|u_0+k|$ is at least $L/32$, independently of the coset offset: the interval $|u| < L/16$ contains at most $L/8 + 1$ test sites and carries less than half of that Green mass. It follows that the original span satisfies

$$\mathrm{span}_i \le K(1 + d|z|v_i).$$

Centering and $\sqrt{v_i} \le H$ give $\sqrt{v_i} \le K + Kd|z|H\sqrt{v_i}$. Since $dH \to 0$, this bounds all v_i and then all spans by a fixed constant. With these improved bounds the last term in (9.2) vanishes uniformly. The score is strictly concave between original support extrema. Two contacts at distance at least two would have an intermediate site strictly above their contact value by the same Green formula, a contradiction. Each nonconstant original therefore has two adjacent support points. □

All active coordinates now have a common diameter. Full globality implies that they also maximize in the restricted common-diameter class at this same horizon. The finite common-diameter classification of Bell [1, Theorem 1.1] says that every such maximizing array is padded IID. Its theorem includes all horizons and has its finite arithmetic premises supplied there. This proves the sequence conclusion of Theorem 1.2.

If no absolute ε_0 existed, choose at horizon N_j an actual non-IID full global with $L < 1/j$. The sequence conclusion would make those same original arrays padded IID eventually, a contradiction. This proves the qualitative threshold statement.

10 A large binary group

As an example of Theorem 1.2, suppose an actual full global array at horizon $M + r_M$ contains $M \to \infty$ active binary laws of a common diameter h_M, with probabilities in a fixed compact $J \subset (0, 1)$. Let

$$R_{\mathrm{rem}} = \sum_{i \text{ outside the group}} \mathbb{E}|X_i|^3.$$

The direct original-price condition

$$R_{\mathrm{rem}} = o(M^{3/2} h_M^3) \tag{10.1}$$

is sufficient. The given group contributes variance at least $c_J M h_M^2$ and price at most KMh_M^3, so

$$L \le \frac{KMh_M^3 + R_{\mathrm{rem}}}{(c_J M h_M^2)^{3/2}} \longrightarrow 0.$$

Theorem 1.2 therefore makes every sufficiently late original array padded IID binary. Its common diameter is h_M, because the original group is active: each remaining coordinate is either the same binary law or zero. No bound on the number of remaining coordinates is needed beyond their stated additive price.

For instance, if all remaining centered laws have support diameter at most $H_M h_M$, then $R_{\mathrm{rem}} \le r_M H_M^3 h_M^3$. Thus $r_M H_M^3 = o(M^{3/2})$ implies (10.1) directly.

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